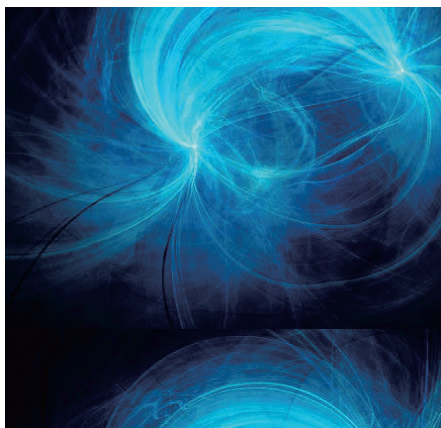


Unit 2: Conductors, Capacitors, Dielectrics

AP Physics C Electricity and Magnetism



Contents

0
1

01 Electrostatics with Conductors

02 Capacitor

TESTDAILY

Electrostatics with Conductors

2.1

TESTDAILY

Property

TESTDAILY

Electrostatic Equilibrium

- Excess charge on an isolated conductor will spread out on the entire conductor until there is **no more movement of the charge**. We say a state of **electrostatic equilibrium** is created on the conductor.
- Property 1:** There is an **absence of electric field inside of the conductor**.
- Property 2:** The excess charge on a conductor **resides entirely on the surface** of the conductor, according to Gauss's Law.

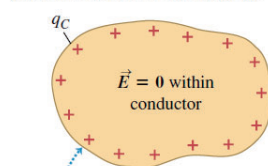
2.1

Property

TESTDAILY

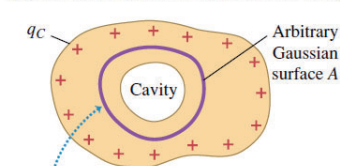
Charge Distribution on a Conductor

(a) Solid conductor with charge q_C



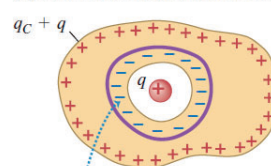
The charge q_C resides entirely on the surface of the conductor. The situation is electrostatic, so $\vec{E} = 0$ within the conductor.

(b) The same conductor with an internal cavity



Because $\vec{E} = 0$ at all points within the conductor, the electric field at all points on the Gaussian surface must be zero.

(c) An isolated charge q placed in the cavity



For $\vec{E}$ to be zero at all points on the Gaussian surface, the surface of the cavity must have a total charge $-q$.

2.1

Exercise

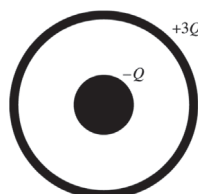
TESTDAILY

2017_Intl_MCQ_27-28

A hollow conducting sphere is surrounded by a larger concentric spherical conducting shell, as shown above. The inner sphere has a net charge of $-Q$, and the outer sphere has a net charge of $+3Q$.

27. What is the net charge on the inner surface of the spherical shell?

- (A) $-Q$
- (B) 0
- (C) $+Q$
- (D) $+2Q$
- (E) $+3Q$



2.1

Exercise

TESTDAILY

2017_Intl_MCQ_27-28

28. What is the net charge on the outer surface of the spherical shell?

- (A) 0
- (B) $+Q$
- (C) $+2Q$
- (D) $+3Q$
- (E) $+4Q$

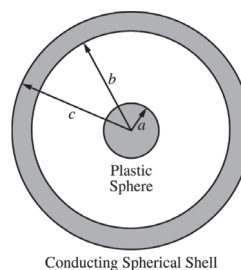
2.1

Example

TESTDAILY

2018_NA_FRQ_1

A solid plastic sphere of radius a and a conducting spherical shell of inner radius b and outer radius c are shown in the figure above. The shell has an unknown charge. The solid plastic sphere has a charge per unit volume given by $\rho(r) = \beta r$, where β is a positive constant and r is the distance from the center of the sphere. Express your answers to parts (a), (b), and (c) in terms of β , r , a , and physical constants, as appropriate.



2.1

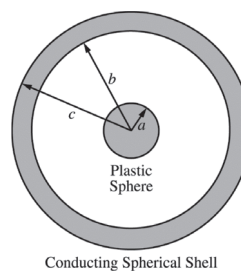
Example

TESTDAILY

2018_NA_FRQ_1

(a) Consider a Gaussian sphere of radius r concentric with the plastic sphere. Derive an expression for the charge enclosed by the Gaussian sphere for the following regions.

- i. $r < a$
- ii. $a < r < b$



2.1

Example

TESTDAILY

2018_NA_FRQ_1

(b) Use Gauss's law to derive an expression for the magnitude of the electric field in the following regions.

i. $r < a$

ii. $a < r < b$

2.1

Example

TESTDAILY

2018_NA_FRQ_1

(c) At any point outside of the conducting shell, it is observed that the magnitude of the electric field is zero.

i. Determine the charge on the inner surface of the conducting shell.

Justify your answer.

ii. Determine the charge on the outer surface of the conducting shell.

2.1

Definition

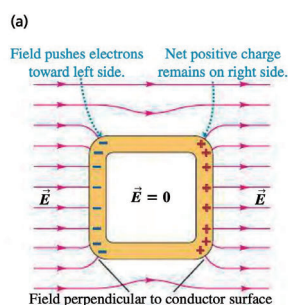
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Electrostatic Shielding (CED p50)

- Electrostatic shielding is the process of surrounding an area by a completely closed conductor to create a region free of an electric field.
- The electric charges within the cage's conducting material are distributed such that they cancel the field's effect in the cage's interior.

Faraday Cage: a grounded metal screen surrounding a piece of equipment to exclude electrostatic and electromagnetic influences.

2.1

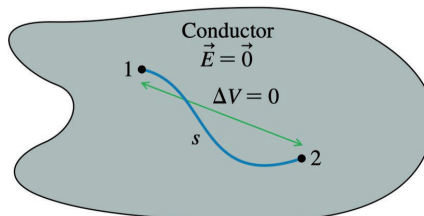


Property

TESTDAILY

Potential of a Conductor

- **Property 3: Equipotential** condition—potential is uniform on a conductor.
- A conducting surface must be an equipotential surface.
- Potential of a conducting sphere $V = k \frac{Q}{R}$



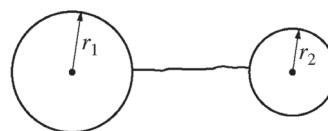
2.1

Exercise

TESTDAILY

2012_Intl_MCQ_14

A metal sphere with radius r_1 has a total electric charge of magnitude q . An uncharged metal sphere with radius r_2 (with $r_1 > r_2$) is then connected by a wire to the first sphere, as illustrated above. The separation of the spheres is much greater than the radius of either sphere.



Note: Figure not drawn to scale.

- When equilibrium is reached, the spheres will have
- charges on their surfaces of equal magnitude and the same sign
 - charges on their surfaces of equal magnitude and opposite sign
 - equal electric fields at their surfaces
 - equal capacitances
 - equal electric potentials

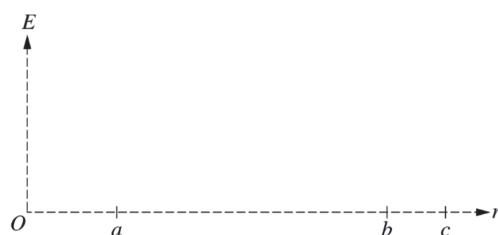
2.1

Example

TESTDAILY

2018_NA_FRQ_1

- (d)
- On the axes below, sketch the electric field E as a function of distance r from the center of the sphere. Sketch the graph for the range $r = 0$ at the center of the sphere to $r = c$ at the outside of the conducting shell.



2.1

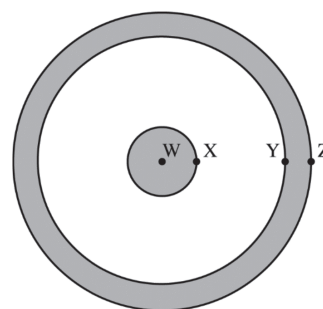
Example

TESTDAILY

2018_NA_FRQ_1

ii. The figure below shows the sphere and shell with four points labeled W, X, Y, and Z. Point W is at the center of the sphere, point X is on the surface of the sphere, and points Y and Z are on the inner and outer surface of the shell, respectively. Rank the points according to the electric potential at that point, with 1 indicating the largest electric potential. If two points have the same electric potential, give them the same numerical ranking.

_____ W _____ X _____ Y _____ Z



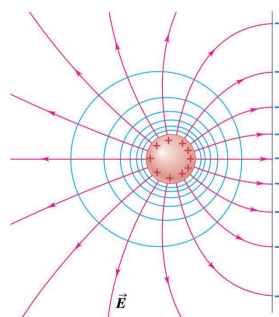
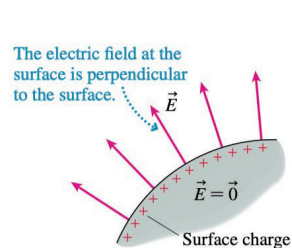
2.1

Property

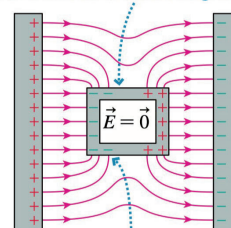
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Field at the Surface of a Conductor

- **Property 4:** The electric field is **always perpendicular to the surface**, no matter what shape the surface may be.



- (b) The conducting box has been polarized and has induced surface charges.



The electric field is perpendicular to all conducting surfaces.

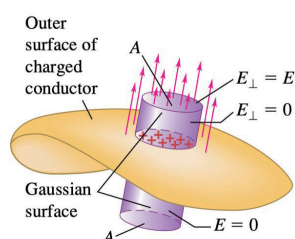
2.1

Property

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Field at the Surface of a Conductor

- **Property 5:** The magnitude of electric field on the surface is proportional to charge density.



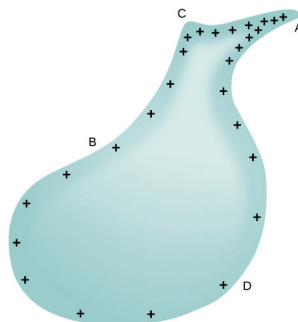
2.1

Property

TESTDAILY

Surface Charge Density

- **Property 6:** Charge density is inversely proportional to the radius of curvature.



2.1

Summary

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Electrostatics with Conductors

1. Charge distribution
2. Electric field inside the conductor
3. Electric potential
4. Electric field on the surface – direction
5. Electric field on the surface – magnitude
6. Charge density

2.1

Capacitor

TESTDAILY



2.2

Capacitor

- 2.2.1

—

Capacitance
- 2.2.2

—

Parallel-Plate Capacitor
- 2.2.3

—

Capacitors of other Geometry
- 2.2.4

—

Energy Storage in Capacitor
- 2.2.5

—

Dielectrics

TESTDAILY

2.2 Capacitor

2.2.1 Capacitance



Concept

TESTDAILY

Capacitor

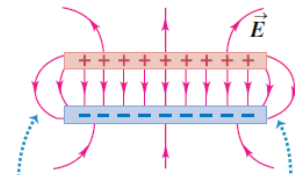
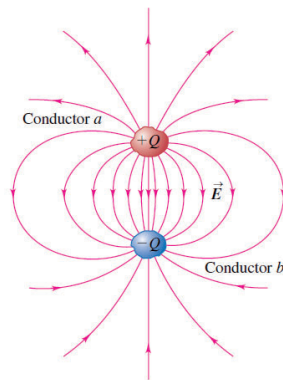
- Any two conductors separated by an insulator (or a vacuum) form a capacitor.
 - They carry charges of same magnitude and opposite signs.
 - Work must be done to create this separation of charge, and, as a result, **potential energy is stored**.
- Capacitors are basically storage devices for electrical potential energy.



Example

Capacitor

TESTDAILY



When the separation of the plates is small compared to their size, the fringing of the field is slight.

2.2.1

Definition

Capacitance (CED p52)

TESTDAILY

Physical quantity	Capacitance 电容
Definition	The ratio of charge to potential difference
Symbol	C
Equation	$C = \frac{Q}{\Delta V}$
Unit	farad, F 法拉 1 F = 1 C/V

- Note: Q is the positive charge quantity, and ΔV is the positive potential difference (voltage) between negative terminal and positive terminal. C is a positive quantity.

2.2.1

2.2 Capacitor

2.2.2 Parallel-Plate Capacitor

2.2.2

Method

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Capacitance of Parallel-Plate Capacitor

Steps in determining capacitance of a capacitor

1. Use Gauss's law to find the strength of the electric field between the plates

$$E = \frac{\sigma}{\epsilon_0}$$

2. Relate the electric field strength to the potential difference between the plates

$$\Delta V = Ed$$

3. Relate the potential difference to capacitance

$$C = \frac{\epsilon_0 A}{d}$$



Example

TESTDAILY

Young_13e_p791_Example24.1

The parallel plates of a 1.0 F capacitor are 1.0 mm apart. What is their area?

Answer: $1.1 \times 10^8 \text{ m}^2$



Exercise

TESTDAILY

Young_13e_p791_Example24.2

The plates of a parallel-plate capacitor in vacuum are 5.00 mm apart and 2.00 m^2 in area. A 10.0 kV potential difference is applied across the capacitor. Compute

- (a) the capacitance;
- (b) the charge on each plate;
- (c) the magnitude of the electric field between the plates.

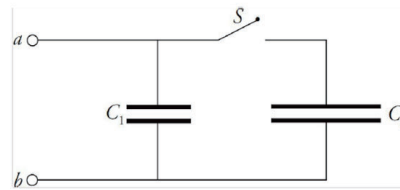
Answer: 3.54 nF; 35.4 μC ; 2 MV/m.



Example

Princeton_2020_Chapter13_Example22

In the diagram below, $C_1 = 2 \text{ mF}$ and $C_2 = 4 \text{ mF}$. When Switch S is open, a battery (which is not shown) is connected between points a and b and charges capacitor C_1 so that $V_{ab} = 12 \text{ V}$. The battery is then disconnected. After the switch is closed, what will be the common potential difference across each of the parallel capacitors (once electrostatic conditions are reestablished)?



Answer: 4 V.

2.2.2

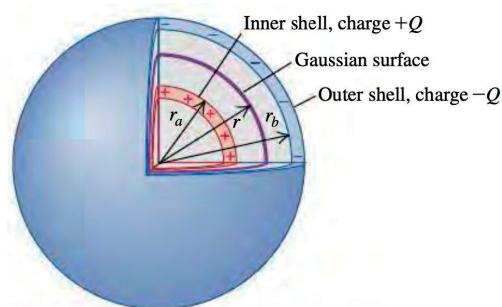
2.2 Capacitor

2.2.3 Capacitors of other Geometry

2.2.3

Property

Spherical Capacitor



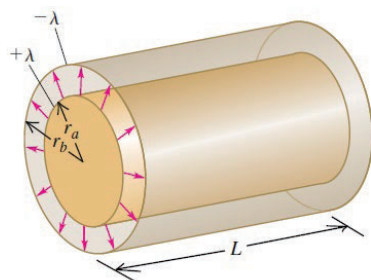
1. Use Gauss's law to find the strength of the electric field between the plates
2. Relate the electric field strength to the potential difference between the plates
3. Relate the potential difference to capacitance

2.2.3

Property

TESTDAILY

Cylindrical Capacitor



1. Use Gauss's law to find the strength of the electric field between the plates
2. Relate the electric field strength to the potential difference between the plates
3. Relate the potential difference to capacitance

2.2.3

2.2 Capacitor

2.2.4 Energy Storage in Capacitor

2.2.4

Property

TESTDAILY

Energy Stored in Capacitor

We can calculate the potential energy of a charged capacitor by calculating the work required to charge it.

$$W = dU = \Delta V dq = \frac{q}{C} dq$$

$$U = \frac{Q^2}{2C} = \frac{1}{2} C (\Delta V)^2 = \frac{1}{2} Q \Delta V$$

2.2.4

Exercise

TESTDAILY

2013_Intl_MCQ_27

Five air-filled parallel-plate capacitors have the plate areas and plate separations listed below, where A and d are constants. The capacitors are each connected to the same potential difference. Which capacitor stores the greatest amount of energy?

- | Area | Separation |
|-----------|------------|
| (A) $2A$ | $d/2$ |
| (B) $2A$ | $2d$ |
| (C) A | d |
| (D) $A/2$ | $d/2$ |
| (E) $A/2$ | d |



Definition

TESTDAILY

Beyond AP: Energy Density (Young 13e p798)

- Energy is stored in the **field** in the region between the plates.
- **Energy density** u is the energy per unit volume in the space between the plates of a parallel-plate capacitor.

$$u = \frac{U}{Ad}$$

$$u = \frac{1}{2} \epsilon_0 E^2$$

(valid for any electric field configuration in vacuum)



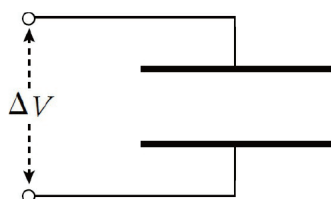
Exam Focus

TESTDAILY

Factors Affecting Parallel-Plate Capacitors

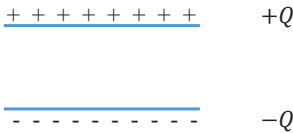
Scenario 1: stay connected with battery -- ΔV is constant

When d or A is changed, then C, Q, E, U will _____?



Factors Affecting Parallel-Plate Capacitors

Scenario 2: disconnected from battery -- Q is constant
When d or A is changed, then C , ΔV , E , U will _____?



2.2 Capacitor

2.2.5 Dielectrics



Dielectric Constant (Young 13e p800)

- Dielectrics, or insulators, are the materials often inserted between the plates of a capacitor to make the capacitance **greater**.

Physical quantity	Dielectric constant 介电常数
Definition	The ratio of capacitance with the dielectric constant to the original capacitance
Symbol	κ
Equation	$\kappa = \frac{C}{C_0}$
Unit	pure number



Example

TESTDAILY

Dielectric Constant

Values of dielectric constant κ at 20°C

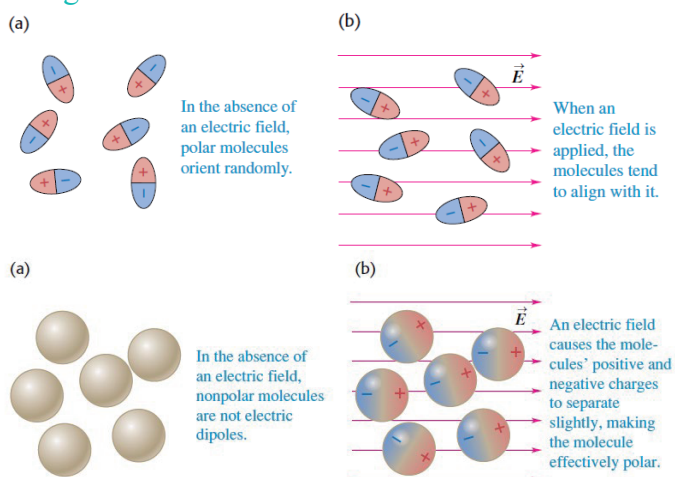
Material	κ	Material	κ
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas	3.40
Air (100 atm)	1.0548	Glass	5-10
Teflon	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3-6	Water	80.4
Mylar	3.1	Strontium titanate	310

2.2.5

Introduction

TESTDAILY

Induced Charge

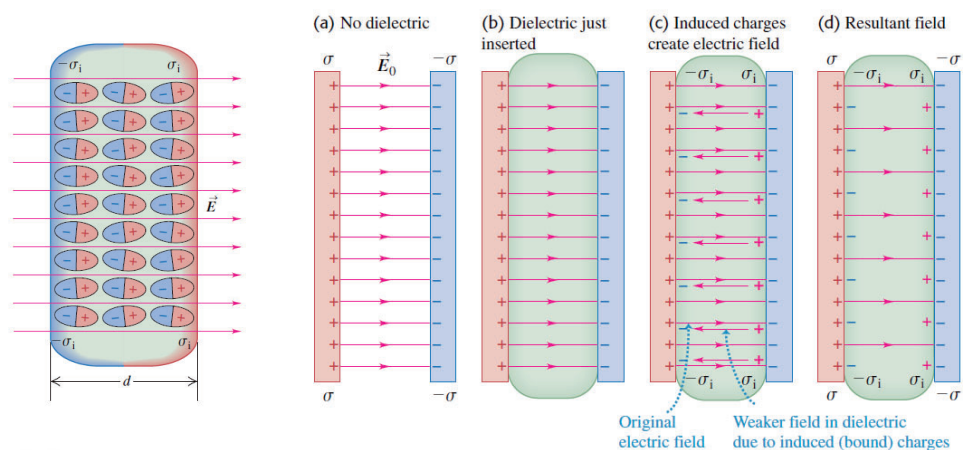


2.2.5

Introduction

TESTDAILY

Induced Charge



2.2.5

Property

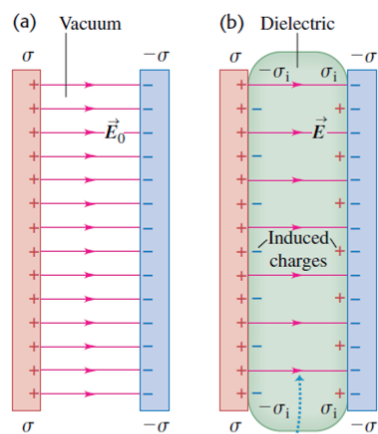
Capacitors with Dielectrics

- $C = \kappa C_0$
- When Q is constant, what is the reduced field?

$$C = \kappa \epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d}$$

ϵ : absolute permittivity of the dielectric

ϵ_0 : permittivity of free space or vacuum



For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

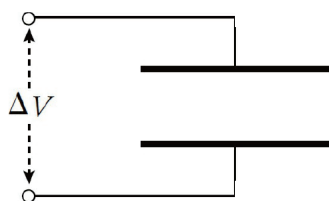
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Exam Focus

Factors Affecting Parallel-plate Capacitors

Scenario 1: stay connected with battery -- ΔV is constant

When a dielectric is inserted, then C, Q, E, U will _____?



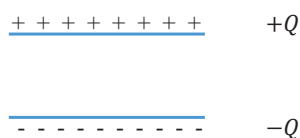
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Exam Focus

Factors Affecting Parallel-Plate Capacitors

Scenario 2: disconnected from battery -- Q is constant

When a dielectric is inserted, then $C, \Delta V, E, U$ will _____?



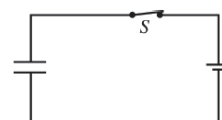
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Exercise

TESTDAILY

2017_Intl_MCQ_11-12

A parallel-plate capacitor connected to a battery is fully charged with the switch S closed, as shown in the circuit above. A slab of dielectric constant $\kappa > 1$ is slowly inserted between the plates of the capacitor.



11. If the switch remains closed when the slab is inserted, what changes, if any, occur?

Potential Difference Across the Plates	Charge on Positive Plate
(A) Remains the same	Increases
(B) Remains the same	Decreases
(C) Increases	Remains the same
(D) Decreases	Increases
(E) Decreases	Remains the same

2.2.5

Exercise

TESTDAILY

2017_Intl_MCQ_11-12

12. If, instead, the switch is open with the capacitor still fully charged when the slab is inserted, what changes, if any, occur?

Potential Difference Across the Plates	Charge on Positive Plate
(A) Remains the same	Increases
(B) Remains the same	Decreases
(C) Increases	Remains the same
(D) Decreases	Increases
(E) Decreases	Remains the same

2.2.5

Example

TESTDAILY

2015_Intl_MCQ_14

If a dielectric is inserted between the plates of a capacitor while the capacitor maintains its connection to a constant voltage source, which of the following is true?

- (A) The capacitance of the capacitor is unchanged.
- (B) The charge on the capacitor plates increases.
- (C) The potential difference across the capacitor increases.
- (D) The electric field between the capacitor plates increases.
- (E) The electric field between the capacitor plates decreases.

2.2.5

Definition

TESTDAILY

Dielectric Strength (Young 13e p805)

- When a dielectric is subjected to a sufficiently strong electric field, **dielectric breakdown** takes place and the dielectric becomes a conductor.
- The maximum electric-field magnitude that a material can withstand without the occurrence of breakdown is called its **dielectric strength**.

Material	Dielectric Constant, K	Dielectric Strength, E_m (V/m)
Polycarbonate	2.8	3×10^7
Polyester	3.3	6×10^7
Polypropylene	2.2	7×10^7
Polystyrene	2.6	2×10^7
Pyrex glass	4.7	1×10^7

2.2.5

Answer to MCQ

TESTDAILY

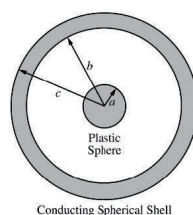
Section	Source	Answer
2.1	2017_Intl_MCQ_27	C
2.1	2017_Intl_MCQ_28	C
2.1	2012_Intl_MCQ_14	E
2.2.4	2013_Intl_MCQ_27	A
2.2.5	2017_Intl_MCQ_11	A
2.2.5	2017_Intl_MCQ_12	E
2.2.5	2015_Intl_MCQ_14	B

2

Answer to FRQ

TESTDAILY

2018_NA_FRQ_1 15 points total

Distribution
of points


A solid plastic sphere of radius a and a conducting spherical shell of inner radius b and outer radius c are shown in the figure above. The shell has an unknown charge. The solid plastic sphere has a charge per unit volume given by $\rho(r) = \beta r$, where β is a positive constant and r is the distance from the center of the sphere. Express your answers to parts (a), (b), and (c) in terms of β , r , a , and physical constants, as appropriate.

2

Answer to FRQ

TESTDAILY

2018_NA_FRQ_1

- (a) Consider a Gaussian sphere of radius r concentric with the plastic sphere. Derive an expression for the charge enclosed by the Gaussian sphere for the following regions.

- i. 3 points

$$r < a$$

For using the integral to determine the charge enclosed	1 point
$q_{enc} = \int \rho dV = \int \beta r dV$	
For correctly substituting for dV into the integration to determine the charge enclosed	1 point
$q_{enc} = \int (4\pi r'^2)(\beta r') dr' = 4\beta\pi \int_{r'=0}^{r'=r} r'^3 dr' = 4\beta\pi \left[\frac{r'^4}{4} \right]_{r'=0}^{r'=r}$	
For a correct answer	1 point
$q_{enc} = \beta\pi r^4$	

2

Answer to FRQ

TESTDAILY

2018_NA_FRQ_1

- ii. 1 points

$$a < r < b$$

For an answer consistent with (a)(i)	1 point
$q_{enc} = \beta\pi a^4$	

2

Answer to FRQ

TESTDAILY

2018_NA_FRQ_1

- (b) Use Gauss's law to derive an expression for the magnitude of the electric field in the following regions.

- i. 2 points

$$r < a$$

For correctly substituting the area of a Gaussian sphere into Gauss's law	1 point
$\frac{q_{enc}}{\epsilon_0} = \oint \vec{E} \cdot d\vec{A}$	
$\frac{q(r)}{\epsilon_0} = EA_{sphere} = E(4\pi r^2)$	
For correctly substituting the charge from part (a)(i) into the equation above	1 point
$\frac{\beta\pi r^4}{\epsilon_0} = E(4\pi r^2)$	
$E = \frac{\beta r^2}{4\epsilon_0}$	

2

Answer to FRQ

TESTDAILY

2018_NA_FRQ_1

ii. 1 point

$a < r < b$

For correctly substituting the charge from part (a)(ii) into Gauss's law to calculate the electric field		1 point
$\frac{\beta\pi a^4}{\epsilon_0} = E(4\pi r^2)$		
$E = \frac{\beta a^4}{4\epsilon_0 r^2}$		

2

Answer to FRQ

TESTDAILY

2018_NA_FRQ_1

(c) At any point outside of the conducting shell, it is observed that the magnitude of the electric field is zero.

i. 2 points

Determine the charge on the inner surface of the conducting shell. Justify your answer.

For an answer consistent with part (a)(ii) (must have opposite sign)		1 point
$q_{inner} = -\beta\pi a^4$		
For a correct justification		1 point
$E_{shell} = 0 = \frac{q_{enc}}{\epsilon_0} \therefore q_{enc} = 0 = q_{sphere} + q_{inner} \therefore q_{inner} = -q_{sphere}$		

ii. 1 point

Determine the charge on the outer surface of the conducting shell.

A correct answer		1 point
$q_{outer} = 0$		

2

Answer to FRQ

TESTDAILY

2018_NA_FRQ_1

(d)

i. 3 points

On the axes below, sketch the electric field E as a function of distance r from the center of the sphere. Sketch the graph for the range $r = 0$ at the center of the sphere to $r = c$ at the outside of the conducting shell.

For concave up graph that begins at the origin and increases to $r = a$		1 point
For a concave up graph that is continuous with the first section and decreases from $r = a$ to nonzero value at $r = b$		1 point
For a graph with a value of $E = 0$ for $r > b$		1 point
Note: Due to scale, discontinuity at point b is not required.		

2

Answer to FRQ

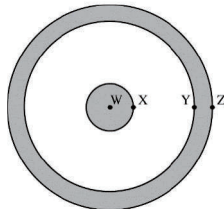
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- (d) (continued)
ii. 2 points

The figure below shows the sphere and shell with four points labeled W, X, Y, and Z. Point W is at the center of the sphere, point X is on the surface of the sphere, and points Y and Z are on the inner and outer surface of the shell, respectively. Rank the points according to the electric potential at that point, with 1 indicating the largest electric potential. If two points have the same electric potential, give them the same numerical ranking.

__1__ W __2__ X __3__ Y __3__ Z



2

Answer to FRQ

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For any ranking that has Y and Z at the same potential	1 point
For ranking the electric potentials: $W > X > Y = Z$ (because $Y = Z$ is the first point)	1 point

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